Inorganic Cume
October 4, 2002

(Do parts 1,3,4 **AND one of 2a OR 2b**)

Metal carbonyl chemistry

1) Give specific examples of each of the following classes of metal – CO species. Your examples must be consistent with the electron counting rules for organometallic complexes.

   a) neutral mononuclear complex with [CO/metal] not equal to five.
   b) binuclear metal carbonyl
   c) carbonyl hydride
   d) carbonylate anion
   e) cyclopentadienyl carbonyl

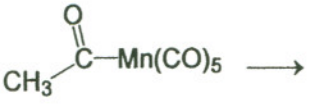
Do either Part 2a **OR** 2b

2a) Give a reasonable synthetic scheme for each of the following:

   a) $CH_3Mn(CO)_5$
   b) $Mn(CO)_5Br$
   c) $Fe_2(CO)_9$ and $Fe_3(CO)_{12}$

2b) Predict the initial products for the following reactions:

   a) $H_2O + Fe(CO)_5 \longrightarrow$

   b) $Me_3N{\rightarrow}O$ +  $\longrightarrow$

3) The binuclear species $Cr_2(CO)_8$ exhibits two isomeric forms. Draw these two structures and suggest how you could easily distinguish between these two forms.

4) Several other ligands have bonding characteristics (i.e. monhapto, pi-acceptor) similar to those of CO. Give the formulas of at least three different ligands of this type. For each indicate the donor atom and the nature of the pi-acceptor orbital.